Automated High-Speed Tape & Reel Packaging Cell

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-TR-001

1. Machine Overview & Manufacturing Context

Equipment Model:	Automated High-Speed Tape & Reel Packaging Cell	**Document ID:**	VAI-MAN-TR-001
Machine Type Key:	tape-reel	**Cleanroom Bay / Stage:**	Bay 6B: Tape & Reel Packaging
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Automated tape and reel packaging system for loading tested IC packages into carrier tape pockets, sealing anti-static cover tape, and winding completed reels for customer shipping.

Process Flow: Tested IC devices are picked from test trays and placed into embossed carrier tape pockets with orientation verified by vision cameras. A constant-temperature floating heat seal bar seals anti-static cover tape over carrier pockets, and finished tape is wound onto 13-inch ESD reels.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
Heat Seal Bar	Applies controlled heat and pressure to seal cover tape over carrier pockets.	Seal bar temp (°C), seal pressure (bar), dwell time (ms).	Temperature fluctuation (>±3°C), heater element resistance drift.
Tape Indexer Sprocket Drive	Indexes carrier tape pocket-by-pocket with high precision.	Indexer vibration (mm/s), step positioning accuracy (µm).	Indexer mechanical vibration (>0.5 mm/s), pocket pitch drift.
Cover Tape Tensioner & Peel Sensor	Maintains uniform tape tension and verifies peel strength compliance.	Tape peel force (N), tensioner spring load (N).	Peel force out of 0.3 - 0.7N EIA-481 spec, tape tearing.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
Heat Seal Bar Temp temp_sealer	°C	168 – 182	**182 – 200**	**200 – 240**	Higher is Worse
Tape Peel Force peel_force	N	0.35 – 0.55	**0.55 – 0.85**	**0.85 – 2**	Higher is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 2,400 operating hours. **Model:** RUL = BaseLife * (1 - (0.45*wear_temp_sealer + 0.35*wear_peel + 0.20*wear_vib))

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
Seal Temp	temp_sealer	°C	**0.45**	175	205	HIGHER IS WORSE
Peel Force	peel_force	N	**0.35**	0.45	0.9	HIGHER IS WORSE
Indexer Vibration	vibration_indexer	mm/s	**0.20**	0.3	0.85	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	**1.00 (100%)**	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Cover Tape Peel Force Out of EIA-481 Spec SYM-TR-001	**HIGH**	temp_sealer, peel_force	• Seal bar temperature drift • Seal pressure misalignment • Cover tape adhesive degradation	Calibrate seal bar temperature and inspect floating pressure springs.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: Cover Tape Seal Bar Temp Fluctuation & Peel Force Variance (SCEN-TR-001)	Severity: WARNING
Telemetry Sensor Pattern:	temp_sealer > 185°C, peel_force > 0.65 N
Possible Root Causes:	Heater thermocouple contact degradation | Adhesive residue on seal bar
Recommended Corrective Action:	Clean brass seal blade and recalibrate temperature controller PID loop.
Verification & Recovery Steps:	Execute 10-meter peel force test on calibrated pull tester (0.35-0.55N verified) > Inspect pocket seal width under optical comparator